

News Weekly

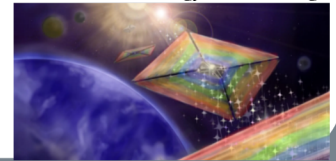
AI ban in NYC schools for young students

New York City schools are set to introduce a new policy restricting the use of artificial intelligence for students and teachers. Starting next week, a one-year pause on AI use will be implemented for students until they reach high school, affecting nearly 600,000 young learners. Additionally, device usage, like laptops and tablets, will be limited, with preschool through 2nd graders barred from using them in class, and 3rd to 8th graders facing screen time limits. Mayor Zohran Mamdani and Schools Chancellor Kamar Samuels will announce these changes, aiming to protect the essential human connection and creativity in education. While teachers retain the ability to use AI for lesson planning, they are restricted from using it for grading or crisis management. The NYC Department of Education has crafted a guidebook with input from various stakeholders to support these measures, making NYC's AI policies among the most stringent in the nation. (NBC)



Interstellar solar sails hit a strange problem at 75% of light speed

The concept of using solar sails propelled by powerful lasers to reach other stars is a promising avenue for interstellar travel. However, recent research by Chao Shen and Jiaze Li highlights a potential complication when these sails travel at relativistic speeds. Initially, the light from the lasers provides thrust by striking the sail, with momentum contributions from incident light, specular reflection, and diffuse scattering. At lower speeds, these forces collectively push the sail forward. But as the sail approaches 75% of the speed of light, the situation becomes more complex due to relativistic effects such as the Doppler shift and light aberration. The Doppler shift reduces the frequency of incoming light, thus decreasing thrust efficiency. More critically, at high speeds, the direction of diffusely scattered light shifts forward, creating a drag effect due to the reaction force. While this doesn't entirely negate the forward thrust, it significantly diminishes the propulsion system's efficiency, posing a challenge to maintaining high speeds with current laser technology. This insight underscores the need for further innovation in propulsion methods to overcome the limitations imposed by relativistic physics. (Science Daily)



Life uses 4 DNA letters. Scientists just made 8 work

Researchers at the University of California San Diego have demonstrated that RNA polymerase, an enzyme essential for gene expression, can accurately read and transcribe DNA containing an expanded genetic alphabet of eight letters. This groundbreaking study combined biochemical experiments with high-resolution cryo-electron microscopy to observe RNA polymerase from *E. coli* bacteria as it interacted with synthetic DNA. The enzyme was able to recognize and incorporate these artificial base pairs, which do not naturally occur, using similar biochemical and structural signals as with natural base pairs. This discovery highlights the enzyme's ability to handle synthetic genetic information, suggesting potential advancements in synthetic biology. In a related study published in PNAS, the team observed RNA polymerase recognizing another set of synthetic base pairs, which lack the typical hydrogen bonds found in DNA. These findings pave the way for future technologies, including new diagnostic tools and therapeutics, by leveraging expanded genetic codes to engineer biological systems with novel functions. (Science Daily)



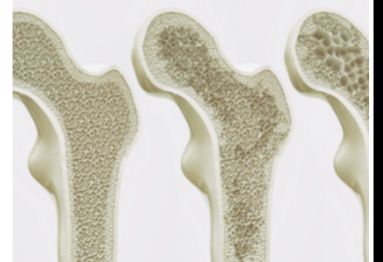
Colorado family wakes up to bear eating breakfast in kitchen

In Colorado Springs, a family awoke to an unexpected visitor—a bear dining in their kitchen. On July 21, security footage captured the bear entering through a back door left open for ventilation. Nerissa England recounted that her curious 8-year-old daughter, Zara Lee, discovered the bear enjoying Flintstones vitamins and rummaging through their trash. Her father quickly joined, and together, they scared the bear away by making noise. "We'll keep the door shut until bears hibernate," England noted, highlighting their newfound caution as they adapt to living alongside local wildlife. (Fox)



Too much coffee may weaken bones, study finds

A recent study from Flinders University, published in *Nutrients*, explored the impact of coffee and tea consumption on bone health in nearly 10,000 women aged 65 and older over a decade. The research focused on bone mineral density (BMD), a crucial indicator of osteoporosis risk. The findings suggest that tea drinkers had slightly higher hip BMD, a difference that could reduce fracture risk across large populations. Conversely, heavy coffee consumption, over five cups daily, was linked to lower BMD, particularly in women with higher alcohol intake. Tea's catechins may aid bone health, while coffee's caffeine might hinder calcium absorption. Moderate coffee intake appeared neutral, especially when balanced with milk. This study highlights the nuanced effects of these popular beverages on aging bone. (Science Daily)



Jokes

What does the sun drink out of?

Sunglasses.

Celebrating lake Xochimilco



	7		3	5		8		
	3	8	7	1	4		6	9
6	4	5				7	1	3
5	8		1			4		
		2			9	3		7
3	9		4	7	8	2	5	1
9	5		2	4	1			
	6		8	9	5	1		2
8	2	1	6	3	7			5

Bacalhau na Frigideira

Ingredients

- - 600 grams cod fillets, cut into 3 cm pieces
- - 150 grams potatoes, peeled and diced
- - 100 grams onion, finely chopped
- - 30 grams garlic, minced
- - 15 grams fresh ginger, grated
- - 10 grams turmeric powder
- - 5 grams cumin powder
- - 400 milliliters coconut milk
- - 100 grams tomato paste
- - 30 grams fresh coriander, chopped

1. Season cod with salt and pepper. In a large pot, sauté onion, garlic, and ginger in a tablespoon of oil until fragrant.
2. Add turmeric and cumin powders, stir for 1 minute until aromatic.
3. Stir in tomato paste, cooking for 2 minutes to deepen flavor.
4. Add diced potatoes and coconut milk. Bring to a simmer, then reduce heat and cook for 10 minutes until potatoes are tender.
5. Gently add cod pieces to the simmering sauce. Cook for 5-7 minutes, or until fish is cooked through and flakes easily.
6. Taste and adjust seasoning as needed.
7. Garnish generously with fresh coriander before serving.
8. Serve with steamed white rice or crusty bread to soak up the delicious sauce.

